K-Medoid Clustering Algorithm

			Cluster 1	Cluster 2	
Point	X	Y	Distance from Centroid A1 (7,2)	Distance from Centroid B1 (5,4)	Assigned Cluster
D1	2	3	5.099019514	3.16227766	C2
D2	5	7	5.385164807	3	C2
D3	8	3	1.414213562	3.16227766	C1
D4	3	5	5	2.236067977	C2
D5	7	2	0	2.828427125	C1
D6	6	8	6.08276253	4.123105626	C2
D7	1	4	6.32455532	4	C2
D8	4	6	5	2.236067977	C2
D9	9	5	3.605551275	4.123105626	C1
D10	5	4	2.828427125	0	C1

			Cluster 1	Cluster 2	
Point	X	Y	Distance from Centroid A1 (1,4)	Distance from Centroid B1 (5,4)	Assigned Cluster
D1	2	3	1.414213562	3.16227766	C1
D2	5	7	5	3	C2
D3	8	3	7.071067812	3.16227766	C2
D4	3	5	2.236067977	2.236067977	C2
D5	7	2	6.32455532	2.828427125	C2
D6	6	8	6.403124237	4.123105626	C2
D7	1	4	0	4	C1
D8	4	6	3.605551275	2.236067977	C2
D9	9	5	8.062257748	4.123105626	C2
D10	5	4	4	0	C2

Euclidian Distance: $\sqrt{|x_{i1}-x_{j1}|^2+|x_{i2}-x_{j2}|^2 \ldots}$

Manhattan Distance: $|x_{i1}-x_{j1}|+|x_{i2}-x_{j2}| \ldots$

Supremum Distance: $\text{Max}\,(|x_{i1}-x_{j1}|+|x_{i2}-x_{j2}| \ldots)$

In this example, we used Euclidian Distance:

	Calculating Cost	
D5(7,2)	**1.41+0+3.6+2.82=**	**7.848191963**
D10(5,4)	**3.1+3+2.2+4.1+4+2.2=**	**18.75751924**
	Total Cost...	**26.6057112**

	Calculating Cost	
D7(1,4)	**1.41+0=**	**1.414213562**
D10(5,4)	**3+3.1+2.23+2.82+4.1+2.2+4.1=**	**21.70905199**
	Total Cost...	**23.12326555**

Swap Cost= New Cost - Previous Cost

= 23.123-26.605 = -3.48244565 <0

As the swap cost is less than zero, we swap the cluster and final medoids are (1,4) and (5,4)